Supporting Information: Supporting Information: Resistance-Resilient Polypharmacology in African Natural-Product-Inspired Antimalarial Space

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This supplement contains the numerical material needed to reproduce the integrated computational analysis. It is organized by evidence layer: chemical space, target-anchored docking, RRS, and exploratory cross-metric analysis. The four-target matrix is reported target by target; no cross-target mean is used because the Vina score is not calibrated across unlike receptor pockets.

S1. Cohort and evidence boundaries

The integrated analysis uses the 17-member P2 Set-C polypharmacology-oriented cohort. Set C was carried forward from the previously defined P2 polypharmacology-oriented cohort after its documented weighted-MPO, synthetic-accessibility, QED, and target-profile filtering; all 17 entries had $n_{\text{targets bound}} = 2$ in the source selection table. V6 introduced no new selection threshold or retrospective re-ranking; this enrichment is intentional and limits prevalence-based interpretation. The upstream $n_{\text{targets bound}} = 2$ label describes the P2 selection criterion; it does not constrain the separate 17-by-4 wild-type docking profile reported here. The docking calculations follow AutoDock Vina conventions.[S1,S2] The MPO top-20 and the P2 MD top-20 are distinct sets. The P2 MD systems 201, 438, the 164-labelled PfClpR cohort, and 214 are not MD validation of Set C. The target panel contains four wild-type docking profiles, whereas RRS is calculated from the separate WT/mutant docking panel and is available only for PfDHFR and PfCRT mutant panels.

S2. Chemical-space coverage and target-anchored docking

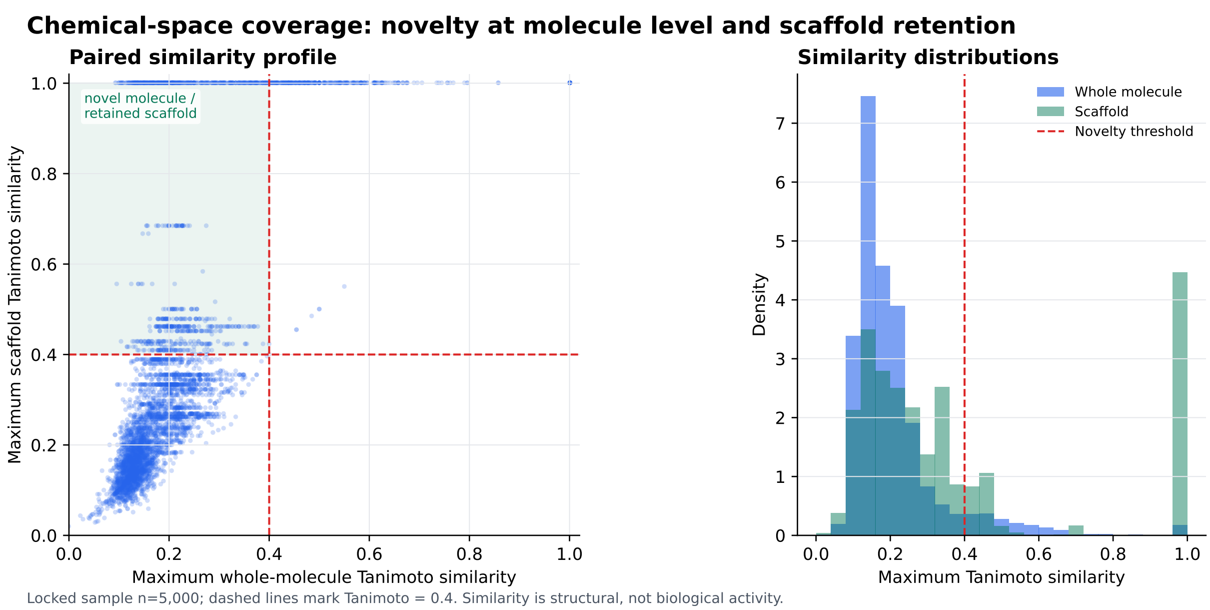

Figure S1: Chemical-space coverage in the locked 5 000-molecule novelty-analysis sample. The paired plot compares maximum whole-molecule and scaffold Morgan-fingerprint Tanimoto similarities to the African-natural-product reference set; the dashed lines mark the operational threshold of 0.4. The upper-left region represents molecules that are dissimilar at the whole-molecule level while retaining scaffold-level similarity. Similarity is a structural descriptor and does not indicate biological activity, target preference, or synthetic success.

S3. Target-anchored docking matrix

Table S1: Target-anchored AutoDock Vina score estimates for the locked, intentionally enriched 17-member Set-C cohort. Values are in kcal mol^{-1}; they are scoring-function outputs, not experimental free energies. Columns are reported target by target because scores from unlike receptor pockets are not calibrated for arithmetic comparison.

Candidate	PfDHFR	PfCRT	PfClpP	PfATP4
PP-01	−5.947	−6.483	−5.467	−6.361
PP-02	−5.562	−6.124	−5.907	−6.591
PP-03	−6.089	−6.633	−6.358	−7.311
PP-04	−5.493	−5.329	−5.386	−6.086
PP-05	−6.285	−6.412	−6.466	−6.378
PP-06	−6.267	−7.907	−7.031	−7.285
PP-07	−6.226	−6.633	−6.226	−5.775
PP-08	−4.935	−5.213	−5.052	−5.494
PP-09	−5.693	−6.871	−5.613	−6.078
PP-10	−6.346	−6.395	−6.016	−6.670
PP-11	−6.179	−6.376	−6.583	−6.724
PP-12	−5.248	−6.357	−5.386	−4.635
PP-13	−5.921	−5.685	−6.454	−7.565
PP-14	−4.863	−5.328	−5.047	−5.460
PP-15	−6.045	−6.084	−6.366	−7.016
PP-16	−5.469	−6.202	−6.203	−6.451
PP-17	−5.266	−5.116	−5.342	−5.360

S4. RRS definition and class summary

For candidate i, mutation m, and target t:

$$\mathrm{RRS}_{i,m,t} = 100|S_{\mathrm{Vina},i,m,t}|/|S_{\mathrm{Vina},i,WT,t}|.$$

Here S_{Vina} denotes an AutoDock Vina scoring-function output, not an experimental free energy. RRS eligibility is determined from the separate WT/mutant docking panel, not from the four-target wild-type matrix. Only genuine wild-type binders with $|S_{\mathrm{Vina},i,WT,t}| \geq 5\,\mathrm{kcal\,mol}^{-1}$ contributed. The 17-candidate class counts were A*: 6, B: 5, C: 5, and D: 1. The observed RRS range was 68.2 to 111.7. The complete machine-readable values are

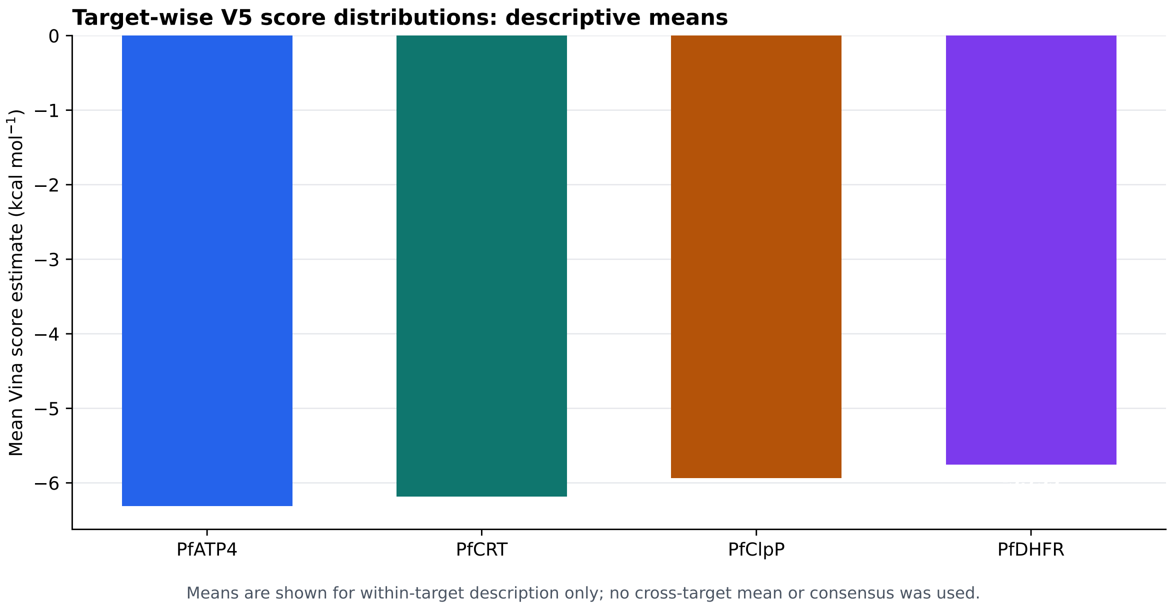

Figure S2: Descriptive target-wise means from the raw docking panel. Bars show the within-target mean across the 17 candidates. Means are shown only within target; no cross-target arithmetic mean or consensus score was used. These summaries are computational scoring-function outputs, not experimental affinity measurements.

provided in the accompanying repository.

Table S2: RRS class definitions used in the integrated analysis.

Class	Operational definition	Number
A*	maximum absolute eligible WT score $\geq 7\,\mathrm{kcal\,mol^{-1}}$ and all available RRS values $\geq 80\,\%$	6
A	All available RRS values $\geq 80\,\%$, without the A* wild-type criterion	0
B	All available RRS values $\geq 70\,\%$, but not all $\geq 80\,\%$	5
C	At least one RRS value $\geq 80\,\%$, with the complete-panel criteria for A/A* or B not met	5
D	No available RRS value $\geq 80\,\%$	1

Table S3: Spearman correlations on the 17-member cohort. Bonferroni-adjusted threshold: $\alpha = 0.017$ for the three hypothesis-oriented comparisons.

Pair	ρ	p	Interpretation
PNS–RRS	−0.559	0.020	Nominal trend; not Bonferroni-significant
ACSI–RRS	−0.132	0.613	Not significant
RRS–wild-type Vina score	−0.433	0.082	Not significant
PNS–wild-type Vina score	0.389	0.123	Descriptive/mechanical comparison

S5. Exploratory cross-metric analysis

S6. Target evidence classes and limitations

The panel spans four distinct antimalarial vulnerabilities: PfDHFR represents folate metabolism and antifolate resistance; PfCRT represents a resistance-linked digestive-vacuole transporter; PfClpP represents parasite proteostasis through the caseinolytic protease, a vulnerability supported by recent work on apicoplast chaperonin–Clp proteostasis;[S3] and PfATP4 represents P-type ATPase-mediated ion regulation, for which an endogenous structure and modulator-bound state have recently been reported.[S4] PfDHFR uses a catalytic-site ligand anchor; PfClpP uses a catalytic triad; PfCRT uses a membrane-associated Y01 cavity proxy; and PfATP4 uses conserved P-type ATPase machinery without a co-crystallized inhibitor. Consequently, all four-target profiles are exploratory and are not equivalent evidence of biological target engagement. No RRS values are reported for PfClpP or PfATP4.

S7. Reproducibility and availability

The integrated candidate manifest, raw docking table, P2 RRS table, cross-metric matrix, scripts, receptor files, ligand files, and provenance records are available in the repository at `https://github.com/NanaEngo/Malaria_codesV2`. The figures and derived tables in this supplement are reproducible computational outputs; automated integrity checks document their derivation but do not substitute for independent structural or experimental validation.

S8. Available machine-readable resources

The accompanying repository provides the computational record used in this study. The available resources include:

- the locked candidate manifest and canonical molecular identifiers;
- the complete target-wise Vina score matrix, ligand poses, and pose-provenance records;
- the declared wild-type/mutant docking panel used for the per-target RRS analysis;
- the derived RRS, PNS, ACSI, and cross-metric tables;
- the chemical-space similarity data and source data for the coverage figure;
- receptor and ligand input files, configuration files, analysis scripts, and figure-generation scripts;
- integrity manifests and reproducibility metadata.

Exact file names, directory paths, checksums, and version-specific dependencies are listed in the repository README and submission manifest rather than in the scientific narrative of this supplement. These resources are intended to support independent computational reproduction; they do not convert docking scores into experimental affinity measurements.

References

(S1) Trott, O.; Olson, A. J. AutoDock Vina: Improving the speed and accuracy of docking with a new scoring function, efficient optimization, and multithreading. *Journal of Computational Chemistry* **2009**, *31*, 455–461.

(S2) Eberhardt, J.; Santos-Martins, D.; Tillack, A. F.; Forli, S. AutoDock Vina 1.2.0: New Docking Methods, Expanded Force Field, and Python Bindings. *Journal of Chemical Information and Modeling* **2021**, *61*, 3891–3898.

(S3) Tissawak, A.; Rosin, Y.; Katz Galay, S.; Qasem, A.; Shahar, M.; Trabelsi, N.; Furman-Schueler, O.; Johnson, S. M.; Florentin, A. A chaperonin complex regulates organelle proteostasis in malaria parasites. *PLOS Pathogens* **2025**, *21*, e1013275.

(S4) Haile, M. T.; Shukla, A.; Zhen, J.; Mather, M. W.; Bhatnagar, S.; Morrisey, J. M.; Zhang, Z.; Vaidya, A. B.; Ho, C.-M. Endogenous structure of antimalarial target PfATP4 reveals an apicomplexan-specific P-type ATPase modulator. *Nature Communications* **2025**, *16*, 9092.